

Matrix Method for Lucas Numbers



lucas.pdf
PDF

read the pdf and explain it to me in details

This paper titled "**On Lucas Numbers by the Matrix Method**" by Fikri Köken and Durmuş Bozkurt presents a **matrix-based approach** to analyze **Lucas numbers**, similar to how Fibonacci numbers are often studied using matrices.

Let's go through it in **detailed and understandable steps**, starting from the basics:

1 2
3 4

1. Background: Fibonacci and Lucas Numbers

- **Fibonacci sequence:**

$$F_0 = 0, F_1 = 1, \text{ and } F_n = F_{n-1} + F_{n-2} \text{ for } n \geq 2$$

- **Lucas sequence:**

$$L_0 = 2, L_1 = 1, \text{ and } L_n = L_{n-1} + L_{n-2}$$

They are related but start with different initial values.

1 2
3 4

2. The Fibonacci Q-Matrix

The **Fibonacci Q-matrix** is defined as:

$$Q = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$$

It has this property:

$$Q^n = \begin{bmatrix} F_{n+1} & F_n \\ F_n & F_{n-1} \end{bmatrix}$$

This allows us to derive identities like:

$$F_{n-1}F_{n+1} - F_n^2 = (-1)^n$$

◆ 3. Defining the Lucas QL-Matrix

They introduce a **new matrix** for Lucas numbers:

$$QL = \begin{bmatrix} 3 & 1 \\ 1 & 2 \end{bmatrix}$$

And they show two important relationships:

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$$\begin{bmatrix} L_{n+1} \\ L_n \end{bmatrix} = QL \cdot \begin{bmatrix} F_n \\ F_{n-1} \end{bmatrix}$$

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$$5 \cdot \begin{bmatrix} F_{n+1} \\ F_n \end{bmatrix} = QL \cdot \begin{bmatrix} L_n \\ L_{n-1} \end{bmatrix}$$

This reveals a tight connection between Fibonacci and Lucas numbers through this matrix.

📐 4. Theorem 2.1 — Powers of the QL Matrix

They prove:

- If **n is even**:

$$QL^n = 5^{n/2} \cdot \begin{bmatrix} F_{n+1} & F_n \\ F_n & F_{n-1} \end{bmatrix}$$

- If **n is odd**:

$$QL^n = 5^{(n-1)/2} \cdot \begin{bmatrix} L_{n+1} & L_n \\ L_n & L_{n-1} \end{bmatrix}$$

This gives us a way to express both Fibonacci and Lucas numbers using **powers of a single matrix**.

5. Theorem 2.2 — Determinant & Cassini Identities

- (i) $\det(QL^n) = 5^n$
- (ii) $F_{n+1}F_{n-1} - F_n^2 = (-1)^n$
- (iii) $L_{n+1}L_{n-1} - L_n^2 = 5(-1)^{n-1}$

These are generalized **Cassini-type identities** using the matrix method.

6. Theorem 2.3 — Binet Formulas

They derive the well-known **closed-form** (Binet) formulas:

- $F_n = \frac{\alpha^n - \beta^n}{\sqrt{5}}$
- $L_n = \alpha^n + \beta^n$

where $\alpha = \frac{1+\sqrt{5}}{2}$, $\beta = \frac{1-\sqrt{5}}{2}$ are the **golden ratio** and its **conjugate**.

They also show how this comes from **diagonalizing** the QL matrix using its eigenvalues:

- Eigenvalues: $\sqrt{5}\alpha$ and $\sqrt{5}\beta$

This connects matrix diagonalization to Fibonacci/Lucas formulas.

7. Theorem 2.4 — Mixed Identities

They present a series of **identities involving sums and differences** of Fibonacci and Lucas numbers:

For all integers m, n :

1. $5F_{m+n} = L_nL_{m+1} + L_{n-1}L_m$
2. $F_{m+n} = F_nF_{m+1} + F_{n-1}F_m$
3. $L_{m+n} = F_{n+1}L_m + F_nL_{m-1}$

$$4. \quad 5F_{m-n} = (-1)^{n-1}(L_m L_{n+1} - L_{m+1} L_n)$$

$$5. \quad F_{m-n} = (-1)^n(F_m F_{n+1} - F_{m+1} F_n)$$

$$6. \quad L_{m-n} = (-1)^{n-1}(F_m L_{n+1} - F_{m+1} L_n)$$

These identities are derived by **multiplying or inverting** the QL matrices and comparing terms.

8. Final Thoughts

The whole paper is a clever exploration of **how matrix algebra can unify and extend Fibonacci and Lucas number theory**.

Benefits of this approach:

- Provides elegant **proofs** of known identities.
 - Makes it easy to **generate new relationships**.
 - Highlights **deep algebraic structures** behind these sequences.
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If you want, I can also create visual aids, code to compute these identities, or step-by-step examples. Want to see how this works with actual numbers (like $n = 5$)?

more clearly and with details please

Absolutely! Let's dive deeper and explain this paper *step by step* like you're learning it from scratch, using **plain language, examples, and small derivations** to make everything clear.

1. What are Fibonacci and Lucas numbers?

Fibonacci Numbers:

These are numbers in a sequence where each number is the sum of the two before it.

Definition:

$$F_0 = 0, \quad F_1 = 1, \quad F_n = F_{n-1} + F_{n-2} \text{ for } n \geq 2$$

Example:

$$F_2 = 1, \quad F_3 = 2, \quad F_4 = 3, \quad F_5 = 5, \quad F_6 = 8, \quad F_7 = 13, \dots$$

◆ Lucas Numbers:

Same rule, but different starting values.

Definition:

$$L_0 = 2, \quad L_1 = 1, \quad L_n = L_{n-1} + L_{n-2}$$

Example:

$$L_2 = 3, \quad L_3 = 4, \quad L_4 = 7, \quad L_5 = 11, \quad L_6 = 18, \quad L_7 = 29, \dots$$

Both sequences follow the **same recurrence relation**, but start differently.



2. The Fibonacci Matrix Method

There's a clever trick: You can use **matrices** to compute Fibonacci numbers.

Define the matrix:

$$Q = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$$

This is called the **Fibonacci Q-matrix**.

It has a powerful property:

$$Q^n = \begin{bmatrix} F_{n+1} & F_n \\ F_n & F_{n-1} \end{bmatrix}$$



What does this mean?

If you raise Q to the power of n , the resulting matrix contains Fibonacci numbers!

Example: Q^3

Let's calculate Q^3 step-by-step:

We already know:

$$Q = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$$

Then:

- $Q^2 = Q \cdot Q = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$
- $Q^3 = Q^2 \cdot Q = \begin{bmatrix} 3 & 2 \\ 2 & 1 \end{bmatrix}$

That gives us:

$$Q^3 = \begin{bmatrix} F_4 & F_3 \\ F_3 & F_2 \end{bmatrix} = \begin{bmatrix} 3 & 2 \\ 2 & 1 \end{bmatrix}$$

✓ It works!

3. Introducing the Lucas QL Matrix

The paper creates a similar matrix for **Lucas numbers**:

$$QL = \begin{bmatrix} 3 & 1 \\ 1 & 2 \end{bmatrix}$$

This new matrix also connects Fibonacci and Lucas numbers in beautiful ways.

 The core relations:

(a) Forward relation:

$$\begin{bmatrix} L_{n+1} \\ L_n \end{bmatrix} = QL \cdot \begin{bmatrix} F_n \\ F_{n-1} \end{bmatrix}$$

✓ This means: You can compute Lucas numbers **from Fibonacci numbers** using this matrix.

(b) Backward relation:

$$5 \cdot \begin{bmatrix} F_{n+1} \\ F_n \end{bmatrix} = QL \cdot \begin{bmatrix} L_n \\ L_{n-1} \end{bmatrix}$$

✓ Similarly, you can compute Fibonacci numbers from Lucas numbers using the same matrix.

Example:

Let's try $n = 2$:

- $F_2 = 1, \quad F_1 = 1$
- Multiply:

$$QL \cdot \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 & 1 \\ 1 & 2 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 4 \\ 3 \end{bmatrix}$$

This gives us:

- $L_3 = 4, \quad L_2 = 3$

✓ Correct again!

4. Theorem 2.1 — Powers of the QL Matrix

This is the main result of the paper:

If n is even:

$$QL^n = 5^{n/2} \cdot \begin{bmatrix} F_{n+1} & F_n \\ F_n & F_{n-1} \end{bmatrix}$$

If n is odd:

$$QL^n = 5^{(n-1)/2} \cdot \begin{bmatrix} L_{n+1} & L_n \\ L_n & L_{n-1} \end{bmatrix}$$

 **Example: Try $n = 2$ (even)**

Then:

$$QL^2 = 5^{2/2} \cdot \begin{bmatrix} F_3 & F_2 \\ F_2 & F_1 \end{bmatrix} = 5 \cdot \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 10 & 5 \\ 5 & 5 \end{bmatrix}$$

Now compute QL^2 directly:

$$QL^2 = QL \cdot QL = \begin{bmatrix} 3 & 1 \\ 1 & 2 \end{bmatrix} \cdot \begin{bmatrix} 3 & 1 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 10 & 5 \\ 5 & 5 \end{bmatrix}$$

 Match confirmed!



5. Determinants and Identities (Theorem 2.2)

(i) Determinant:

$$\det(QL^n) = 5^n$$

This is a clean formula saying that as you raise QL to a power, its determinant grows as a power of 5.

(ii) Cassini's Identity for Fibonacci:

$$F_{n+1}F_{n-1} - F_n^2 = (-1)^n$$

✓ This is a classic identity — it shows how Fibonacci numbers are tightly related.

(iii) Cassini's Identity for Lucas:

$$L_{n+1}L_{n-1} - L_n^2 = 5(-1)^{n-1}$$

This is a **Lucas version** of the identity, with a 5 multiplier because Lucas numbers grow faster.



6. Binet Formulas (Theorem 2.3)

The authors derive **closed formulas** using square roots, which let you compute any term directly.

◆ For Fibonacci:

$$F_n = \frac{\alpha^n - \beta^n}{\sqrt{5}}$$

◆ For Lucas:

$$L_n = \alpha^n + \beta^n$$

Where:

- $\alpha = \frac{1+\sqrt{5}}{2} \approx 1.618$ (golden ratio)
 - $\beta = \frac{1-\sqrt{5}}{2} \approx -0.618$
-

7. Identities Between F and L (Theorem 2.4)

These are some advanced but very useful identities combining Fibonacci and Lucas numbers.

Let's see a few clearly:

 Identity 1:

$$5F_{m+n} = L_n L_{m+1} + L_{n-1} L_m$$

 Identity 2:

$$F_{m+n} = F_n F_{m+1} + F_{n-1} F_m$$

 Identity 3:

$$L_{m+n} = F_{n+1} L_m + F_n L_{m-1}$$

These are useful in proofs and calculations.

Summary

The paper shows how:

- **Matrices** can be used to compute both Fibonacci and Lucas numbers.
 - There's a special **QL matrix** for Lucas numbers.
 - Powers of this matrix give **exact formulas** for F_n and L_n .
 - Many famous identities can be **proved easily** using this matrix method.
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If you'd like, I can write a Python code snippet that computes Fibonacci/Lucas numbers using these matrix methods — or even help you visualize this as a chart or animation.

Want to go further with this?

from this, show me the way to use diagonal matrix to calculate the formula for determining lucas sequence

Great! Let's go step by step to **derive the formula for Lucas numbers** using **diagonalization** of a matrix — a key idea in linear algebra.

We'll show how **Lucas numbers** can be calculated using **powers of a diagonal matrix**, which is much easier to compute than regular matrix powers.

Overview: What We're Doing

We'll:

1. Use the **Lucas matrix** $QL = \begin{bmatrix} 3 & 1 \\ 1 & 2 \end{bmatrix}$
 2. **Diagonalize it**: turn it into a diagonal matrix using eigenvalues and eigenvectors
 3. Use this to compute QL^n , then derive the **Lucas number formula**.
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Step 1: Eigenvalues of the QL Matrix

We want to find the **eigenvalues** λ of:

$$QL = \begin{bmatrix} 3 & 1 \\ 1 & 2 \end{bmatrix}$$

We solve the **characteristic equation**:

$$\begin{aligned} \det(QL - \lambda I) &= 0 \\ \begin{vmatrix} 3 - \lambda & 1 \\ 1 & 2 - \lambda \end{vmatrix} &= (3 - \lambda)(2 - \lambda) - 1 = 0 \\ &= \lambda^2 - 5\lambda + 5 = 0 \end{aligned}$$

Solve using quadratic formula:

$$\lambda = \frac{5 \pm \sqrt{25 - 20}}{2} = \frac{5 \pm \sqrt{5}}{2}$$

So we have:

- $\lambda_1 = \sqrt{5} \cdot \alpha$
- $\lambda_2 = \sqrt{5} \cdot \beta$

where:

- $\alpha = \frac{1+\sqrt{5}}{2}$
 - $\beta = \frac{1-\sqrt{5}}{2}$
-



Step 2: Diagonalization

If a matrix is diagonalizable, then:

$$QL = PDP^{-1}$$

Where:

- D is the **diagonal matrix** with eigenvalues
- P is the matrix of eigenvectors

Then:

$$QL^n = PD^nP^{-1}$$

This makes computing powers of QL super easy since D^n is just raising each diagonal entry to the power n .



Step 3: Apply to Lucas Numbers

From the paper (Theorem 2.1):

If n is **odd**, then:

$$QL^n = 5^{(n-1)/2} \cdot \begin{bmatrix} L_{n+1} & L_n \\ L_n & L_{n-1} \end{bmatrix}$$

So, using diagonalization:

$$\begin{bmatrix} L_{n+1} & L_n \\ L_n & L_{n-1} \end{bmatrix} = \frac{1}{5^{(n-1)/2}} \cdot P D^n P^{-1}$$

This is how the **Lucas numbers are embedded in the matrix** you get from diagonalizing QL^n .

✨ Final Step: Extracting the Lucas Formula

Using the eigenvalues:

- $\lambda_1 = \sqrt{5} \cdot \alpha$
- $\lambda_2 = \sqrt{5} \cdot \beta$

We find that:

$$L_n = \alpha^n + \beta^n$$

This is the **Binet formula for Lucas numbers**, and it comes from diagonalizing QL , because the eigenvalues encode the growth of the Lucas sequence.

✅ Summary

To use diagonalization to compute **Lucas numbers**:

1. Use the matrix $QL = \begin{bmatrix} 3 & 1 \\ 1 & 2 \end{bmatrix}$
2. Find eigenvalues: $\lambda_1 = \sqrt{5} \cdot \alpha, \lambda_2 = \sqrt{5} \cdot \beta$
3. Diagonalize: $QL = PDP^{-1}$, so $QL^n = PD^n P^{-1}$
4. Relate QL^n to Lucas terms:

For **odd** n ,

$$QL^n = 5^{(n-1)/2} \cdot \begin{bmatrix} L_{n+1} & L_n \\ L_n & L_{n-1} \end{bmatrix}$$

5. The eigenvalue growth leads to the clean formula:

$$L_n = \alpha^n + \beta^n$$

Would you like to see a **step-by-step numeric example** for, say, L_5 ? Or should I write a Python function to compute it using diagonalization?